

The Gaussian Moments Conjecture in Two Variables

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Abstract

The Gaussian Moments Conjecture asks whether a polynomial whose positive powers all have Gaussian integral zero must eventually remain zero after multiplication by any fixed polynomial. We prove the conjecture for two real Gaussian variables. The proof uses circular coordinates to separate rotation from radius. The angular part becomes a constant-term problem, while the radial part becomes the simple functional $U^j \mapsto j!$. An exposed lower face of the resulting Newton diagram, the constant-term theorem of Duistermaat and van der Kallen, and reduction modulo a prime then isolate a nonzero lowest term. This forces every polynomial with vanishing pure moments to have rotational weights of one strict sign, from which the mixed-moment conclusion is immediate. Together with the recent counterexamples of Long in every dimension at least three, this determines the exact dimensional range of the conjecture.

1 Introduction

Let $X_1, \dots, X_d$ be independent standard real Gaussian random variables. The Gaussian Moments Conjecture, introduced by Derksen, van den Essen, and Zhao [1], is the following statement.

Gaussian Moments Conjecture GMC(d). If $P \in \mathbb{C}[X_1, \dots, X_d]$ satisfies

$$\mathbb{E}(P^m) = 0 \quad \text{for every } m \geq 1,$$

then, for every $Q \in \mathbb{C}[X_1, \dots, X_d]$,

$$\mathbb{E}(QP^m) = 0 \quad \text{for all sufficiently large } m.$$

The conjecture was motivated in part by its connection with the Jacobian Conjecture [1]. Long recently found explicit counterexamples to GMC(d) for every $d \geq 3$ [3]. The two-variable case was therefore the last undecided dimension.

Theorem 1.1. *The Gaussian Moments Conjecture holds in two real variables.*

The proof is short once the right coordinates are chosen. In two dimensions, rotation has a single integer weight and there is a single radial variable. Thus every polynomial can be displayed schematically as

$$P = \sum_k \underbrace{T^k}_{\text{rotation}} \underbrace{B_k(U)}_{\text{radius}}.$$

Gaussian integration first takes the constant term in T , then replaces U^j by $j!$. If both positive and negative rotational weights occur, the lowest radial orders have a supporting line above weight

zero. The polynomial carried by that lower face has a nonzero constant term in some power, by a theorem of Duistermaat and van der Kallen [2]. Raising that power to a suitable prime and reducing modulo the prime makes the lowest term impossible to cancel after the factorial substitution. This contradicts the assumed moment vanishing.

The argument is conceptual rather than computational. In particular, it contains the previously studied three-level and star-shaped supports without needing separate case analyses.

2 Circular coordinates

Let X, Y be independent standard real Gaussian variables. Set

$$Z = \frac{X + iY}{\sqrt{2}}, \quad W = \frac{X - iY}{\sqrt{2}}.$$

Wick's formula gives

$$\mathbb{E}(Z^a W^b) = \begin{cases} a!, & a = b, \\ 0, & a \neq b. \end{cases} \quad (1)$$

Introduce a radial variable U and a Laurent variable T through the injective substitution

$$Z \mapsto T, \quad W \mapsto UT^{-1}.$$

Thus $Z^a W^b$ becomes $T^{a-b} U^b$. Every polynomial $P(Z, W)$ therefore has a unique expression

$$P = \sum_{k \in S} T^k B_k(U), \quad (2)$$

where $S \subset \mathbb{Z}$ is finite and every $B_k \in \mathbb{C}[U]$ is nonzero. The integer k is the rotational weight.

Define the factorial functional

$$\mathcal{L} : \mathbb{C}[U] \longrightarrow \mathbb{C}, \quad \mathcal{L}(U^j) = j!.$$

Equation (1) is exactly the identity

$$\mathbb{E}(F) = \mathcal{L}(\text{CT}_T F) \quad (F \in \mathbb{C}[Z, W]). \quad (3)$$

This is the feature special to two variables: there is one angular constant-term operation and one radial factorial.

3 The exposed lower face

For each $k \in S$, write

$$\nu_k = \text{ord}_U B_k, \quad b_k = [U^{\nu_k}] B_k \neq 0.$$

The points (k, ν_k) record the smallest radial degree at each rotational weight. We now extract the lower face visible from weight zero.

Lemma 3.1 (Lower face). *Suppose $0 \in \text{conv}(S)$. There are rational numbers ρ, θ , a subset*

$$S_0 = \{k \in S : \nu_k = \rho + \theta k\},$$

and an integer $r \geq 1$ with the following properties:

$$\nu_k \geq \rho + \theta k \quad (k \in S), \quad 0 \in \text{conv}(S_0),$$

and, for

$$P_0(T) = \sum_{k \in S_0} b_k T^k, \quad F_m(U) = \text{CT}_T(P^m),$$

there are $n \in \mathbb{Z}_{\geq 0}$ and $c \in \mathbb{C}^\times$ such that

$$\text{every term of } F_m \text{ has degree at least } \rho m, \tag{4}$$

$$n = \rho r, \quad F_r(U) = cU^n + \text{terms of degree greater than } n. \tag{5}$$

Proof. Minimize

$$\sum_{k \in S} \lambda_k \nu_k$$

subject to

$$\lambda_k \geq 0, \quad \sum_k \lambda_k = 1, \quad \sum_k k \lambda_k = 0.$$

The feasible set is nonempty because $0 \in \text{conv}(S)$. Let the minimum be ρ . Elementary linear-programming duality supplies $\theta \in \mathbb{Q}$ such that

$$\nu_k \geq \rho + \theta k.$$

An optimal solution is supported on the equality set S_0 , so $0 \in \text{conv}(S_0)$.

We use the constant-term theorem of Duistermaat and van der Kallen [2]: if every positive power of a complex Laurent polynomial has zero constant term, then zero does not lie in the convex hull of its support. Applied contrapositively to P_0 , it gives an $r \geq 1$ for which

$$c := \text{CT}_T(P_0^r) \neq 0.$$

Consider a weight-zero product of m terms selected from (2). If the selected weights are $k_1, \dots, k_m$, then its radial degree is at least

$$\sum_{a=1}^m \nu_{k_a} \geq \rho m + \theta \sum_{a=1}^m k_a = \rho m.$$

This proves (4). When $m = r$, equality can occur only by taking the lowest coefficient from weights on S_0 , and the sum of all equality terms is c . Hence the lowest term of F_r is $cU^{\rho r}$. Its exponent is necessarily a nonnegative integer; call it n . $\square$

4 Prime isolation

The next lemma is the arithmetic step. Its role is simple: after a prime dilation, Frobenius preserves only p -th powers, while the factorial functional kills every radial level above the lowest one.

Lemma 4.1 (Prime-factorial isolation). *Let P and F_m be as in Lemma 3.1. If*

$$\mathcal{L}(F_m) = 0 \quad (m \geq 1),$$

then the conclusion (5) is impossible.

Proof. Let $R \subset \mathbb{C}$ be the $\mathbb{Z}$ -algebra generated by the coefficients of P and by c^{-1} . It is a finitely generated domain of characteristic zero. For all but finitely many rational primes p , there is a homomorphism from R to a field of characteristic p . Indeed, the finite-type morphism $\text{Spec } R \rightarrow \text{Spec } \mathbb{Z}$ is dominant, and its image is constructible by Chevalley's theorem [4]. Since it contains the generic point, it contains a nonempty open subset [5]. Because c is invertible in R , its image remains nonzero under every such homomorphism.

Fix one of these primes. Before reducing modulo p , work in the characteristic-zero domain R . Lemma 3.1 says that every term of F_{rp} has degree at least $\rho rp = np$. Hence $\mathcal{L}(F_{rp}) = 0$ can be written in R as

$$0 = (np)! \sum_{d \geq np} [U^d] F_{rp} \frac{d!}{(np)!}.$$

Since R is a domain of characteristic zero, $(np)! \neq 0$, so it may be cancelled to give

$$\sum_{d \geq np} [U^d] F_{rp} \frac{d!}{(np)!} = 0. \quad (6)$$

We now apply a homomorphism $R \rightarrow \kappa$, where κ is a field of characteristic p , and continue to use the same notation for the reduced polynomials.

In characteristic p , Frobenius and constant-term extraction give

$$F_{rp} = \text{CT}_T((P^r)^p) = (\text{CT}_T(P^r))^p = F_r^p. \quad (7)$$

Indeed, if $H = \sum_i h_i(U)T^i$, then $H^p = \sum_i h_i(U)^p T^{pi}$ in characteristic p ; because $p > 0$, the equality $pi = 0$ holds exactly when $i = 0$. Thus $\text{CT}_T(H^p) = \text{CT}_T(H)^p$. By (5), the coefficient of U^{np} in (7) is c^p . A term whose degree is not divisible by p has zero coefficient modulo p . A remaining term has degree jp for some $j > n$, and its multiplier in (6) is

$$\frac{(jp)!}{(np)!}.$$

This integer is divisible by p , since its product contains $(n+1)p$. Consequently, reducing (6) modulo p leaves only

$$c^p = 0,$$

contrary to the choice of the reduction. □

5 Proof of the theorem

Proof of Theorem 1.1. Write P as in (2) and suppose that $\mathbb{E}(P^m) = 0$ for every $m \geq 1$. By (3), this says

$$\mathcal{L}(F_m) = 0 \quad (m \geq 1).$$

If $0 \in \text{conv}(S)$, Lemma 3.1 produces a nonzero lowest coefficient and Lemma 4.1 gives a contradiction. Therefore

$$0 \notin \text{conv}(S).$$

Since $S \subset \mathbb{Z}$, either every weight of P is strictly positive or every weight is strictly negative.

Now fix a polynomial Q . Its rotational weights form a bounded finite set. Every weight of P^m , on the other hand, tends uniformly toward the same strict side as $m \rightarrow \infty$. Hence QP^m has no weight-zero term once m is sufficiently large:

$$\text{CT}_T(QP^m) = 0.$$

Equation (3) then gives $\mathbb{E}(QP^m) = 0$, as required. □

Corollary 5.1. *The Gaussian Moments Conjecture holds in dimension d exactly when $d \leq 2$.*

Proof. The case $d = 1$ is known [1, Proposition 4.2]. Theorem 1.1 proves the case $d = 2$. Long’s examples disprove the conjecture for every $d \geq 3$ [3]. $\square$

6 What the argument says

The proof gives a stronger structural statement than the mixed-moment conclusion alone:

$$\mathbb{E}(P^m) = 0 \text{ for all } m \geq 1 \implies \text{all rotational weights of } P \text{ have one strict sign.}$$

Thus vanishing in two variables is not caused by a complicated balance among positive, zero, and negative angular modes. Such a balance always has a lowest radial face, and a prime moment detects it.

This also explains why support-by-support calculations quickly appeared to show rigidity. A three-level support, a star support, or a tree of primitive weight relations is only a finite presentation of the same lower face. Leaf removal is unnecessary: the supporting line exposes all relevant leaves at once.

The proof is genuinely two-dimensional. With one circular pair there is one weight coordinate and one radial order, and the latter is measured by the single factorial $j!$. In higher dimension there are several radial directions, so the one-dimensional isolation step no longer applies. Corollary 5.1 shows that this is not merely a technical gap: counterexamples already occur in dimension three.

7 Formal verification

A companion Lean 4 development has been prepared as supplementary material for this paper. It mechanically checks the factorial-quotient divisibility, prime-dilated factorial isolation, commutation of Frobenius with angular constant-term extraction, the final prime contradiction, and the eventual vanishing argument for both signs of the rotational support. The development contains no `sorry` declarations.

The formalization exposes three external inputs as named axioms: the Duistermaat–van der Kallen constant-term theorem, rational supporting-face extraction, and the existence of good finite-characteristic specializations of finite-type $\mathbb{Z}$ -domains. Thus the artifact gives a machine-checked verification of the arithmetic and constant-term core of the proof, with its remaining trust boundary stated explicitly.

Acknowledgments and declaration of generative AI use

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References

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